

Neural Particle Method

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Abstract

1 Introduction

2 Problem Formulation

3 Neural Particle Method

Algorithm 1: Neural Particle Method

Input: Initial particles $\{x_0^{(i)}\}_{i=1}^N$; network b_θ ; step Δt ; number of steps K

Output: Particle trajectories $\{x_k^{(i)}\}_{k=0}^K$

```
1 for  $k = 0, 1, \dots, K - 1$  do
2   for  $i = 1, \dots, N$  do
3     Evaluate  $b_\theta(x_k^{(i)}[k], \{x_k^{(i)}[k]\}_{j=1}^N)$ ;
4     Update  $x_k^{(i)}[k + 1] \leftarrow x_k^{(i)}[k] + \Delta t b_\theta(\cdot) + \sigma \sqrt{\Delta t} \xi_k^{(i)}$ ;
5 return  $\{x_k^{(i)}\}$ ;
```

4 Assumptions

Assumption 1. _____

5 Main Results

Theorem 5.1 (Descriptive title, e.g. propagation of chaos / convergence rate). _____

Proof. _____ $\square$

6 Numerical Experiments

7 Conclusion

150–250
word
abstract.
State:
the
problem
being
solved
(a
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discreti-
sation of
a PDE /
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evolu-
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with a
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method
(neural
particle
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signifi-
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relative
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existing

Table 1: [TODO: Comparison against baseline methods.]

Method	[TODO: Metric 1]	[TODO: Metric 2]	[TODO: Metric 3]
[TODO: Baseline]	—	—	—
[TODO: Neural particle method]	—	—	—

References

- [1] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, Baltimore, MD, 2013.
- [2] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.